

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
5th Semester of B. Pharm. Examination
University Theory Examination November 2017
PH316 Medicinal Chemistry- I

Date: 21.11.2017 , Tuesday

Time: 10.00 a.m. to 1.00 p.m.

Maximum Marks: 80

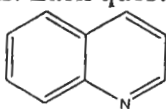
Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

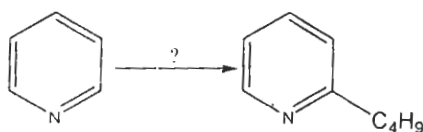
Q 1 Attempt all questions. Each question is of one mark.

20



Which heterocyclic ring is?

1. [A] Isoquinoline
[B] Quinoline
[C] Indole
[D] Isoindole



2. [A] C_4H_9Cl
[B] C_4H_9OH
[C] C_4H_9Li
[D] $C_4H_9OC_4H_9$

-N-C-N- ring system is observed in _____

3. [A] Imidazole
[B] Pyrazole
[C] Pyrrolidine
[D] Pyrrole

In the Gattermann reaction of an aromatic or heteroaromatic compound, _____ of the ring occurs.

4. [A] Formylation
[B] Nitration
[C] Hydroxylation
[D] Carboxylation

Conjugation of a drug includes the following EXCEPT:

5. [A] Glucuronidation
[B] Sulphate formation
[C] Hydrolysis

[D] Methylation

Which of the followings is the second messenger of G-protein-coupled receptor?

6. [A] Adenylyl cyclase
[B] Sodium ions
[C] Phospholipase C
[D] cAMP

Irreversible interaction of an antagonist with a receptor is due to:

7. [A] Ionic bonds
[B] Hydrogen bonds
[C] Covalent bonds
[D] All of the above

Which of the following cholinomimetics is commonly used in the treatment of glaucoma?

8. [A] Pilocarpine
[B] Lobeline
[C] Acetylcholine
[D] Neostigmine

Which of the following drugs is both a muscarinic and nicotinic blocker?

9. [A] Atropine
[B] Benztropine
[C] Hexamethonium
[D] Succinylcholine

Which heterocyclic ring is present in structure of pilocarpine?

10. [A] Benzimidazole
[B] Pyrrole
[C] Imidazole
[D] Benzopyrrole

Which of the following is NOT an insecticide?

11. [A] Chlorpyrifos
[B] Schradan
[C] Parathion
[D] Pralidoxime

Pancuronium acts by:

12. [A] Blocking Muscarinic receptor
[B] Activating Muscarinic receptor
[C] Blocking Nicotinic receptor
[D] Activating Nicotinic receptor

Propranolol is a _____ class of adrenergic antagonist.

13. [A] Aryl ethanolamine
[B] Aryl propanolamine
[C] Aryloxyethanolamine
[D] Aryloxypropanolamine

Prazosin is...

14. [A] Selective α_1 agonist
[B] Selective α_1 antagonist
[C] Nonselective α agonist
[D] Nonselective α antagonist

Stimulation of adrenergic receptors in lung produces...

15. [A] Bronchoconstriction
[B] Bronchodilation
[C] Enhance bronchial secretion
[D] reduce bronchial secretion

Theophylline helps in treatment of asthma by...

16. [A] Antagonizing adenosine
[B] Inhibiting Phosphodiesterase

- [C] Stimulating release of catecholamines
[D] All of above
- Zileuton produce pharmacological effect by...
17. [A] Antagonizing LT receptors
[B] Preventing release of LT from mast cells
[C] Blocking biosynthesis of LT
[D] All of above
- Which of the following ligands does not bind to the cell surface receptors?
18. [A] Interferon
[B] Insulin
[C] Progesterone
[D] Adrenaline
- Ester linkage of cholinomimetics is involved in _____ bond formation at active site.
19. [A] H-Bond
[B] Ionic bond
[C] Hydrophobic Bond
[D] Covalent Bond
- Reactive metabolite of Paracetamol is generated by
20. [A] Glucuronidation
[B] Sulphate Conjugation
[C] Quinon formation
[D] Glutathion Conjugation

SECTION – II

Q 2 Attempt any TWO of the following

- A Add a note on Fischer Indole Synthesis. 05
- B Draw and explain structure of GPCR. Describe mechanism of its activation and signal transduction in brief. 05
- C Write down SAR of Glucocorticoids. 05

Q 3 Attempt any TWO of the following

- A Classify anti asthmatic drugs and write mechanism of action for each class of therapeutic agents. 05
- B Classify acetylcholine esterase inhibitors and explain aging of the enzyme AChE. 05
- C Compare and contrast various structural requirements for a molecule to be adrenergic agonist and antagonist. 05

SECTION – III

Q 4 Attempt any FOUR of the following

- A Write a note on “Cytochrome P450 catalyzed oxidative metabolism”. 05
- B Comment on the followings: 05
1. Imidazole is more basic than pyrazole.
 2. Pyridine undergoes EAS at 2nd position.
- C Give various reactions of furan. 05
- D Add a note on Proton Pump Inhibitors. 05

- E** Explain various factors affecting metabolism of xenobiotics. **05**
- F** Write down the synthetic pathway for Epinephrine and Pentoprazole. **05**

Q 5 Attempt any FOUR of the following

- A** Explain any two methods for preparation of quinolone. **05**
- B** Describe the SAR of Choline esters. **05**
- C** Give various reactions of indole. **05**
- D** Draw the chemical structure of Propranolol, Ephedrine and Salbutamol and write down the synthesis of any one of them. **05**
- E** Classify antiulcerative agents and explain the role of H₁ receptor antagonists. **05**
- F** Explain with suitable example reductive metabolism of drugs. **05**
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